

MUGDHA MEDA

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EDUCATION

Carnegie Mellon University

M.S. Mechanical Engineering (Research) - Robotics, Dynamics & Controls; GPA: 4.0/4.0

Pittsburgh, PA

Expected May 2027

Indian Institute of Technology Madras

B.Tech. Mechanical Engineering; GPA: 9.04/10.0

Chennai, India

May 2025

Coursework: Optimal Control & Reinforcement Learning, Robot Dynamics & Analysis, Advanced Control System Integration, Robot Learning, Reduced Models in Dynamical Systems, AI/ML for Engineers, Modern Control Theory, Theory of Mechanisms, Control for Automotive Systems

TECHNICAL SKILLS

Simulation & Tools: MATLAB/Simulink, ADAMS/View, Ansys IDE (Workbench, Structural), ABAQUS CAE, Webots, Gazebo, Git, Linux

Programming: C, C++, Python, MATLAB, ROS2, Arduino **CAD:** Onshape, SolidWorks, Autodesk Fusion 360, AutoCAD

RESEARCH EXPERIENCE

Graduate Research Assistant - Microrobotics Lab

Carnegie Mellon University

Legged & aerial robot control, state estimation, and hardware build at the centimeter scale (Guide: Dr. Sarah Bergbreiter)

Jan 2026 – Present

- Built a **hybrid stance-flight dynamics model** for a 35 g passive-legged hopping quadcopter and implemented **predictive control** from empirically identified stance-phase return maps to stabilize continuous hopping, **max rmse of 0.1m** while tracking reference trajectories.
- Training an **off-policy RL agent** in a custom simulation environment to capture unmodeled dynamics like re-orientation thrust, drag, and extend hopping to long-horizon, energy-efficient trajectories.
- Designed a **current-sensing-only stance/swing detection MLP Classifier**, engineering features to achieve **0.93+ Balanced Accuracy for an 8 cm quadruped**, to enable real-time gait-phase estimation and control **without encoders**.
- Developed a **phase-based gait controller** that regulates stride frequency and touchdown timing using per-leg PD control. Validated coordinated locomotion in MuJoCo and on hardware, while identifying **sim-to-real discrepancies and classifier failure modes** at varying gait frequencies.
- Developing a **learned mapping** between a **bio-inspired magnetic whisker sensor** and onboard flight EKF to **3D relative airflow velocity**, supervised by motion capture, decoupling ambient wind from drone ego-motion for outdoor wind estimation.

PROFESSIONAL & TEACHING EXPERIENCE

R&D Engineering Intern

Thryv Mobility (Assistive Devices Startup)

Motor control and real-time simulation for adaptive exercise equipment

Jun 2024 – Aug 2024

- Developed **VESC brake-current control algorithms** that modulate motor torque as a function of joint position, velocity, and acceleration to deliver programmable resistance for wheelchair-user strength training.
- Built a real-time **hardware-in-the-loop simulation** of the rig on ESP32 + VESC using **micro-ROS**, bridging embedded firmware to ROS2 and Gazebo for controller validation ahead of hardware trials.
- Designed an adjustable encoder mount absorbing test-rig assembly tolerances, recovering reliable position feedback for closed-loop control.

Graduate Teaching Assistant - Robot Dynamics; Advanced Control Systems Integration

Carnegie Mellon University

Two semesters; weekly recitations, office hours and lab support for 70+ graduate students

Sep 2025 – Present

- Teaching recitations, office hours to clear doubts. Debugged **quadcopter, self-balancing robot, double rotary-pendulum** lab platforms.

Mechanical Lead Engineer - Team Abhiyaan (Autonomous Vehicle Team)

IIT Madras

Led a 5-engineer subteam; \$2K+ budget; IGVC international competition; Autonomous Shuttle Build

Apr 2022 – Oct 2023

- **Owned design, fabrication, and assembly** of an autonomous ground vehicle for an international competition, IGVC'23. Developed wishbone-based **suspension** for the drive and castor wheels, reducing vertical shock amplitudes by **55%**.
- Troubleshot on-site failures at competition to keep the vehicle race-ready. Recruited, trained a **team of 5** for the IGVC'24.
- Built a **Steer-by-Wire** module for an **autonomous shuttle from an e-golf cart** with FEA mounts for rack actuation and mapped steer control.

SELECTED PROJECTS

Full-Stack Autonomous Vehicle Control Stack - Modern Control Theory, CMU

Oct 2025 – Dec 2025

- Implemented **PID, full-state feedback**, and **LQR** lateral/longitudinal controllers on a linearized bicycle model, cutting lap time and trajectory-tracking error in **Webots** closed-loop simulation.
- Closed the loop without default sensing by implementing **EKF SLAM** for localization and **A*** for global path planning.

Dynamic Modeling for Swash Plates in Rotorcrafts - Aerospace Engineering, IIT Madras

Nov 2024 – Mar 2025

- Developed a **high-fidelity ADAMS/View model** of helicopter blade and swash plate dynamics to capture **collective and cyclic pitch control** and blade flap response under varying aerodynamic loads.
- Performed **analytical validation in MATLAB** by evaluating blade flap response across varying **lift and drag loads**, establishing agreement between the dynamic model and expected physical behavior.

Variable Ride-Height Rover Suspension & Articulation (Indian Patent 202341083243)

Sep 2022 – Jan 2024

- Designed an actuated variable ride-height suspension that shifts the rover Center of Gravity in-motion for **+50%** ground clearance, **+30%** gradeability with respect to regular Martian rovers and includes a Chebyshev-linkage articulation system that eliminates bogie overturning.

AWARDS & HONORS

- **1st Place**, AutoNav Design Competition, **Intelligent Ground Vehicle Competition (IGVC) 2023**, Oakland University, MI.
- **Indian Patent**, for a Variable Ride-Height Rover Suspension & Articulation System. (Indian Patent 202341083243).
- **Young Research Fellowship (2023)**, IIT Madras - 1 of 28 selected from 270 applicants.